

Numerical Linear Algebra First-Year Exam, August 2023

Do 4 (four) problems.

1. Suppose A is Hermitian positive definite.

- (a) Prove that each principal submatrix of A is Hermitian positive definite.
- (b) Prove that an element of A with largest magnitude lies on the diagonal.
- (c) Prove that A has a Cholesky decomposition.

2. **Warning: Suspected error.** The source specifies a potentially rectangular matrix as a projector, although projectors and orthogonal projectors ordinarily require a square matrix. In addition, the stated equivalence has the zero orthogonal projector as a possible exception unless nonzero projectors are assumed. The intended dimensions and convention are not uniquely determined, so the source wording has been retained.

Let $P \in \mathbb{C}^{m \times n}$ be a projector. Show that $\|P\|_2 = 1$ if and only if P is an orthogonal projector.

3. Let $A \in \mathbb{C}^{m \times n}$, with $m \geq n$ and $\text{rank}(A) = p = n \geq 3$. Let $a_1, a_2, \dots$ denote the columns of A .

- (a) Using the modified Gram–Schmidt process, write out expressions for q_1, q_2, q_3 , the first three columns of Q in the QR decomposition of A .
- (b) Show the vector q_3 found in part (a) is orthogonal to both q_1 and q_2 .

4. **Warning: Suspected error.** Part (b) refers to n singular values and asserts $\sigma_n > 0$, which requires full column rank and hence $m \geq n$. The condition $n < m$ appears only in part (a), so it is unclear whether it is intended to continue into part (b). The source wording has been retained.

Let $A = U\Sigma V^*$ be the singular value decomposition of $A \in \mathbb{C}^{m \times n}$. Let u_j denote column j of U .

- (a) Suppose $\text{rank}(A) = p < n < m$. Show $\{u_1, u_2, \dots, u_p\}$ is a basis for $\text{Col}(A)$ and $\{u_{p+1}, u_{p+2}, \dots, u_m\}$ is a basis for $\text{Null}(A^*)$.
- (b) Suppose A is full rank and $x \neq 0$. Let $\sigma_i, i = 1, \dots, n$, be the singular values of A . Show

$$\sigma_1 \geq \frac{\|Ax\|_2}{\|x\|_2} \geq \sigma_n > 0.$$

If you want to use the property that $\|A\|_2 = \sigma_1$, then you must prove that.

5. Let $A \in \mathbb{C}^{m \times m}$ be Hermitian.

- (a) Show that all eigenvalues of A are real.
- (b) Define the stationary iterative method (a.k.a. fixed point method)

$$x^{(k+1)} = Ax^{(k)} + b. \tag{1}$$

Suppose (1) has fixed-point x , namely x satisfies $x = Ax + b$. Show the iteration (1) converges to x from any starting guess $x^{(0)}$, that is $x^{(k)} \rightarrow x$ as $k \rightarrow \infty$, if and only if the eigenvalues λ_i of A satisfy $|\lambda_i| < 1, i = 1, \dots, m$. You may use the fact that Hermitian matrix A is unitarily diagonalizable.